Automated QA audit · generated 2026-10-19 12:11

Does the application behave the way its specification says?

Target `http://localhost:8080` · Spec `examples/demo-bank/plan.json` · Run time 102s · Mode replay of saved plan (plan.json)

Northwind Demo Bank: a synthetic banking dashboard with accounts, transfers between own accounts, and a filterable transaction list.

REQUIREMENTS
11
7 pass · 4 fail · 0 unverified

SCENARIOS RUN
16
8 normal · 5 negative · 3 edge

DEFECTS FOUND
5
5 high

LLM COST THIS RUN
$0.0000
0 calls · 0 from cache

COST PER SCENARIO
$0.0000
budget cap $1.00

1 · Requirement verdicts

REQ	REQUIREMENT	VERDICT	BASIS	SCENARIOS
REQ-01 high	**Enter Demo opens dashboard** Clicking Enter Demo on the landing page opens the dashboard without login. PRD: “Clicking it opens the dashboard. No login is required.”	PASSED	all 1 scenarios passed (normal)	SC-01-a
REQ-02 medium	**Accounts shown with formatted balances** Each account shows name, masked number and a USD balance with two decimals. PRD: “balance formatted as US dollars with two decimals”	PASSED	all 1 scenarios passed (normal)	SC-02-a
REQ-03 low	**Main navigation** Navigation offers Dashboard, Transfer and Transactions. PRD: “Navigation offers Dashboard, Transfer and Transactions”	PASSED	all 1 scenarios passed (normal)	SC-03-a

REQ	REQUIREMENT	VERDICT	BASIS	SCENARIOS
REQ-04 high	**Transfer moves exact amount** A successful transfer decreases the source and increases the destination by exactly the amount and shows a reference. PRD: “increases by exactly the transfer amount”	FAILED	1 of 2 scenarios failed: Cents are transferred exactly (10.55)	SC-04-a, SC-04-b
REQ-05 high	**Transfer creates debit and credit records** Each successful transfer adds a debit row for the source and a credit row for the destination. PRD: “creates a debit record … and a credit record”	PASSED	all 1 scenarios passed (normal)	SC-05-a
REQ-06 high	**Amount must be positive with max 2 decimals** Non-numeric, zero, negative or >2-decimal amounts are rejected with an error and no balance change. PRD: “a number greater than zero with at most two decimal places”	FAILED	2 of 4 scenarios failed: Negative amount is rejected; Three decimal places are rejected (1.005)	SC-06-a, SC-06-b, SC-06-c, SC-06-d
REQ-07 high	**Insufficient funds rejected** A transfer above the source balance is rejected with an 'Insufficient funds' message. PRD: “rejected with an "Insufficient funds" message”	FAILED	1 of 1 scenarios failed: Overdraft is rejected with Insufficient funds	SC-07-a
REQ-08 medium	**From and To must differ** Transfers to the same account are rejected. PRD: “From and To accounts must be different”	PASSED	all 1 scenarios passed (negative)	SC-08-a
REQ-09 medium	**Empty amount validation** Submitting without an amount shows a validation message. PRD: “An empty amount shows a validation message”	PASSED	all 1 scenarios passed (negative)	SC-09-a
REQ-10 medium	**Transactions newest first** Transactions are listed newest first with date, account, description, type, amount. PRD: “listed newest first”	PASSED	all 1 scenarios passed (normal)	SC-10-a
REQ-11 high	**Type filter shows only selected type**	FAILED	1 of 2 scenarios failed: Debit filter shows only debits	SC-11-a, SC-11-b

REQ	REQUIREMENT	VERDICT	BASIS	SCENARIOS
	Filtering by Credit or Debit shows only rows of that type. PRD: “shows only transactions of the selected type”			

2 · Defects (5)

HIGH BUG-01 · Not met: Cents are transferred exactly (10.55)

SC-04-b · edge scenario · requirements REQ-04 · reproduced on 2 attempt(s)

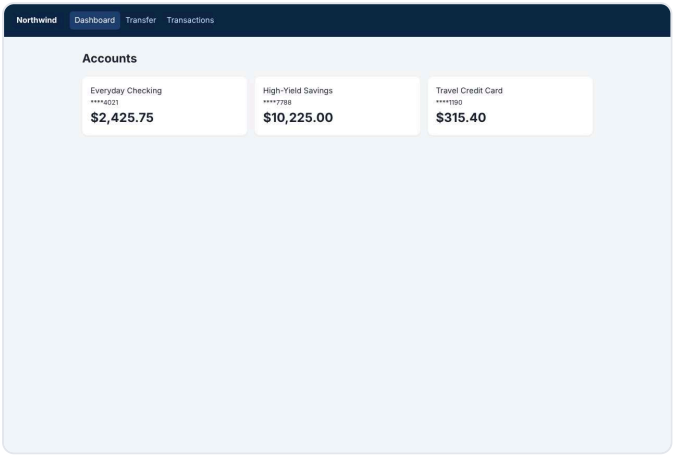
Steps to reproduce

1. Open http://localhost:8080 and click 'Enter Demo'
2. Click 'Transfer' in the top navigation
3. Read 'testid=balance-chk' as chk_before → read 2425.75
4. Read 'testid=balance-sav' as sav_before → read 10225.0
5. Select 'Everyday Checking (****4021)' in 'label=From account'
6. Select 'High-Yield Savings (****7788)' in 'label=To account'
7. Enter '10.55' into 'label=Amount'
8. Click the 'Transfer' submit button
9. Read 'testid=balance-chk' as chk_after → read 2415.2
10. Read 'testid=balance-sav' as sav_after → read 10235.0

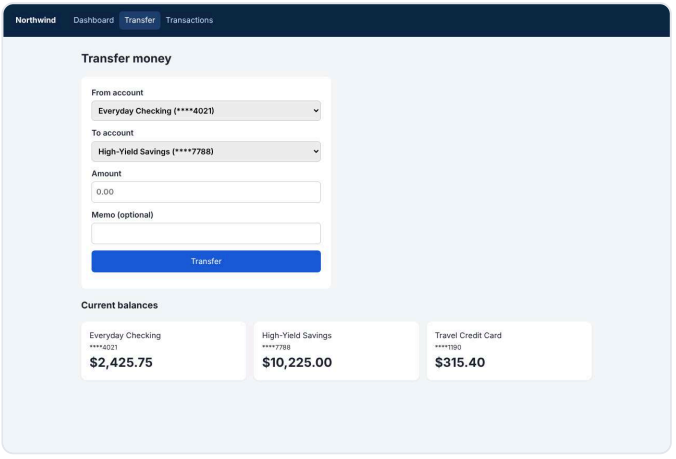
Expected	Both balances move by exactly 10.55.
Observed	sav_after=10235.0, sav_before=10225.0
Failed check	Destination increased by 10.55 expected `{sav_after} == {sav_before} + 10.55` observed `sav_after=10235.0, sav_before=10225.0`
Captured values	`{'chk_before': 2425.75, 'sav_before': 10225.0, 'chk_after': 2415.2, 'sav_after': 10235.0}`

API evidence (1)

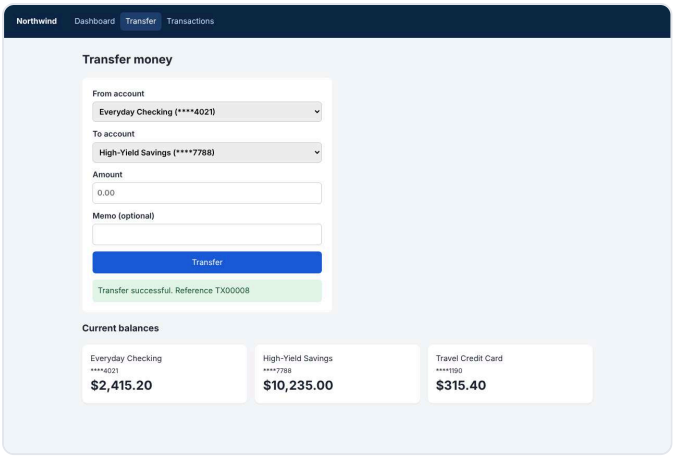
```
POST http://localhost:8080/api/transfer → 200
request: {"from":"chk","to":"sav","amount":"10.55","memo":""}
response: {"ok": true, "reference": "TX00008"}
```



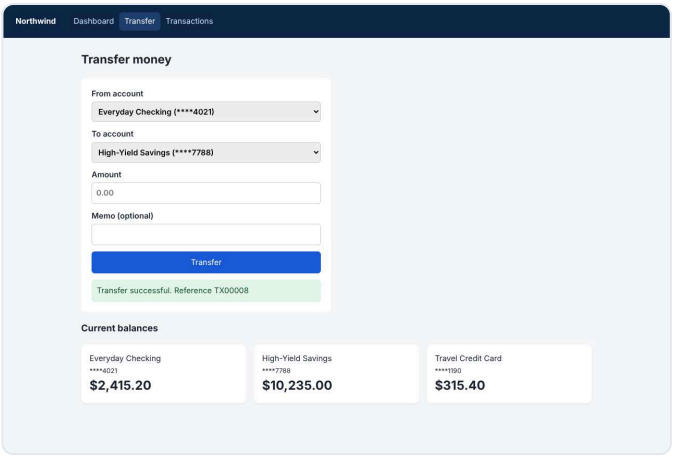
after step 0: Open http://localhost:8080 and click 'Enter Demo'



after step 1: Click 'Transfer' in the top navigation



after step 7: Click the 'Transfer' submit button



after step 9: Read 'testid=balance-sav' as sav_after

HIGH BUG-02 · Not met: Negative amount is rejected

SC-06-a · negative scenario · requirements REQ-06 · reproduced on 2 attempt(s)

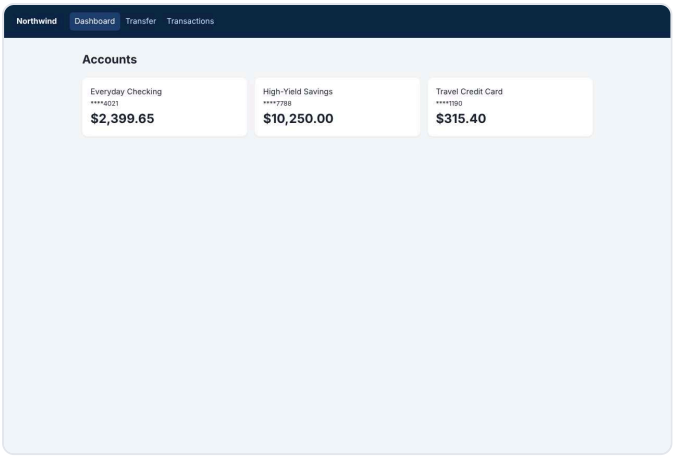
Steps to reproduce

1. Open http://localhost:8080 and click 'Enter Demo'
2. Click 'Transfer' in the top navigation
3. Read 'testid=balance-chk' as chk_before → read 2399.65
4. Read 'testid=balance-sav' as sav_before → read 10250.0
5. Select 'Everyday Checking (****4021)' in 'label=From account'
6. Select 'High-Yield Savings (****7788)' in 'label=To account'
7. Enter '-50' into 'label=Amount'
8. Click the 'Transfer' submit button
9. Read 'testid=balance-chk' as chk_after → read 2449.65
10. Read 'testid=balance-sav' as sav_after → read 10200.0

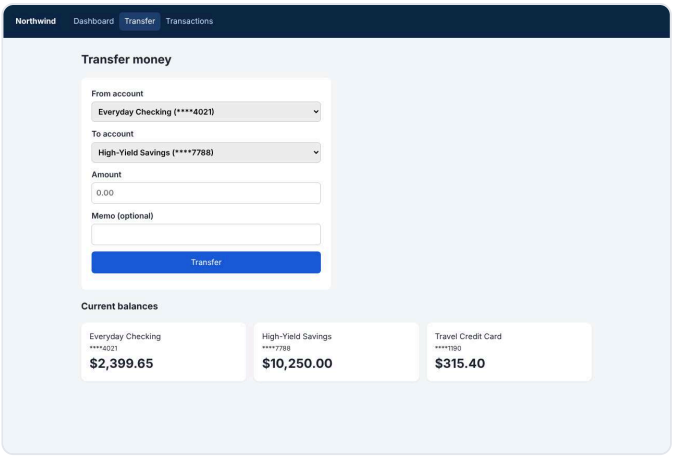
Expected	Error shown, balances unchanged.
Observed	no error or validation message visible; chk_after=2449.65, chk_before=2399.65; sav_after=10200.0, sav_before=10250.0
Failed check	An error message is shown expected `a visible validation / error message` observed `no error or validation message visible`
Failed check	Checking balance unchanged expected `{chk_after} == {chk_before}` observed `chk_after=2449.65, chk_before=2399.65`
Failed check	Savings balance unchanged expected `{sav_after} == {sav_before}` observed `sav_after=10200.0, sav_before=10250.0`
Captured values	`{'chk_before': 2399.65, 'sav_before': 10250.0, 'chk_after': 2449.65, 'sav_after': 10200.0}`

API evidence (1)

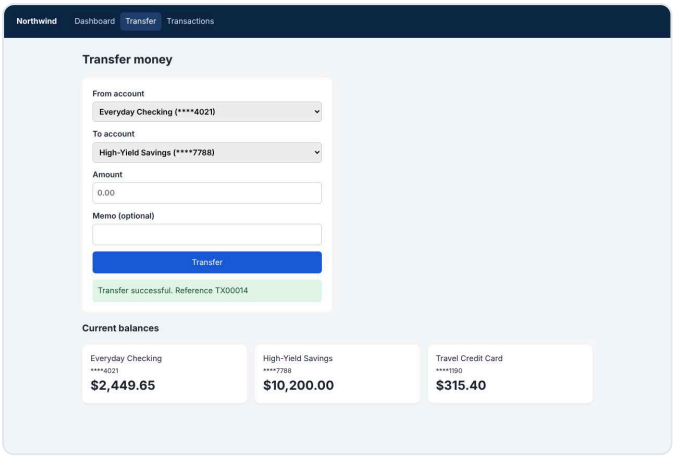
```
POST http://localhost:8080/api/transfer → 200
request: {"from":"chk","to":"sav","amount":"-50","memo":""}
response: {"ok": true, "reference": "TX00014"}
```



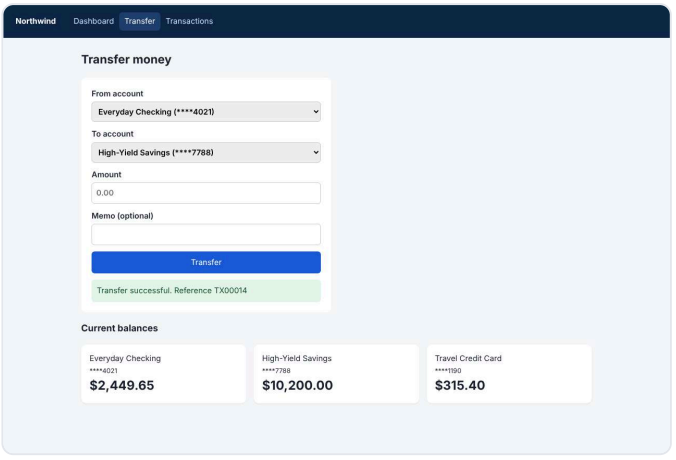
after step 0: Open http://localhost:8080 and click 'Enter Demo'



after step 1: Click 'Transfer' in the top navigation



after step 7: Click the 'Transfer' submit button



after step 9: Read 'testid=balance-sav' as sav_after

HIGH BUG-03 · Not met: Three decimal places are rejected (1.005)

SC-06-d · edge scenario · requirements REQ-06 · reproduced on 2 attempt(s)

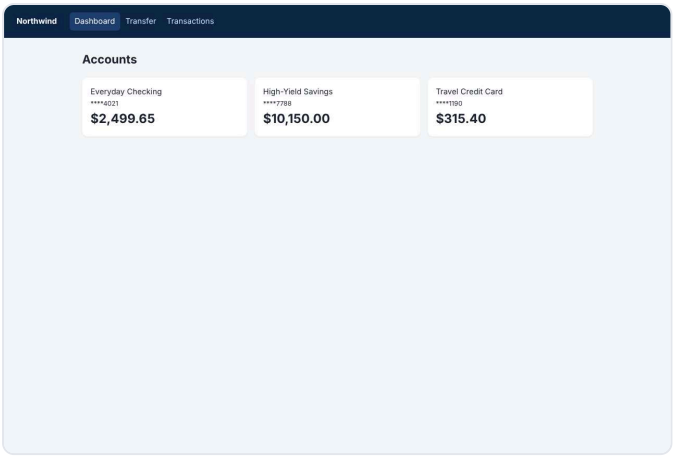
Steps to reproduce

1. Open http://localhost:8080 and click 'Enter Demo'
2. Click 'Transfer' in the top navigation
3. Read 'testid=balance-chk' as chk_before → read 2499.65
4. Read 'testid=balance-sav' as sav_before → read 10150.0
5. Select 'Everyday Checking (****4021)' in 'label=From account'
6. Select 'High-Yield Savings (****7788)' in 'label=To account'
7. Enter '1.005' into 'label=Amount'
8. Click the 'Transfer' submit button
9. Read 'testid=balance-chk' as chk_after → read 2498.64
10. Read 'testid=balance-sav' as sav_after → read 10151.0

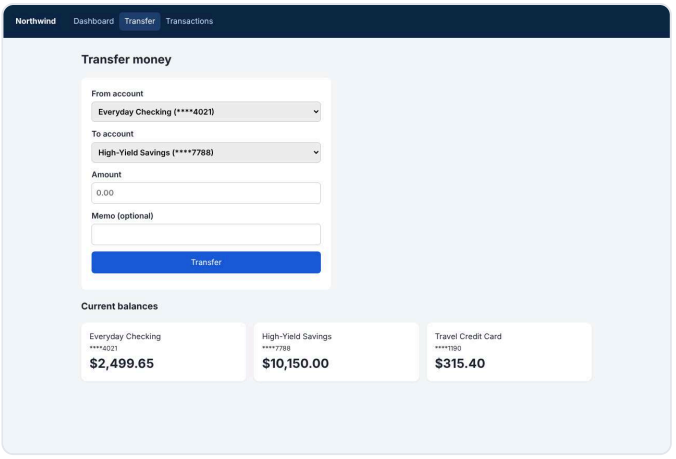
Expected	Error shown, balances unchanged.
Observed	no error or validation message visible; chk_after=2498.64, chk_before=2499.65; sav_after=10151.0, sav_before=10150.0
Failed check	An error message is shown expected `a visible validation / error message` observed `no error or validation message visible`
Failed check	Checking balance unchanged expected `{chk_after} == {chk_before}` observed `chk_after=2498.64, chk_before=2499.65`
Failed check	Savings balance unchanged expected `{sav_after} == {sav_before}` observed `sav_after=10151.0, sav_before=10150.0`
Captured values	`{'chk_before': 2499.65, 'sav_before': 10150.0, 'chk_after': 2498.64, 'sav_after': 10151.0}`

API evidence (1)

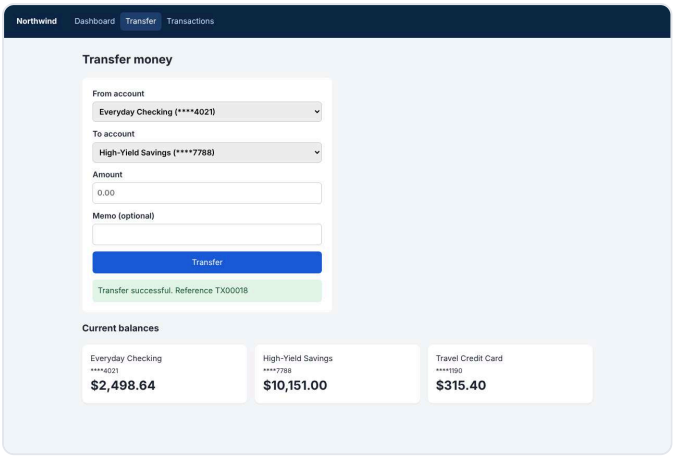
```
POST http://localhost:8080/api/transfer → 200
request: {"from":"chk","to":"sav","amount":"1.005","memo":""}
response: {"ok": true, "reference": "TX00018"}
```



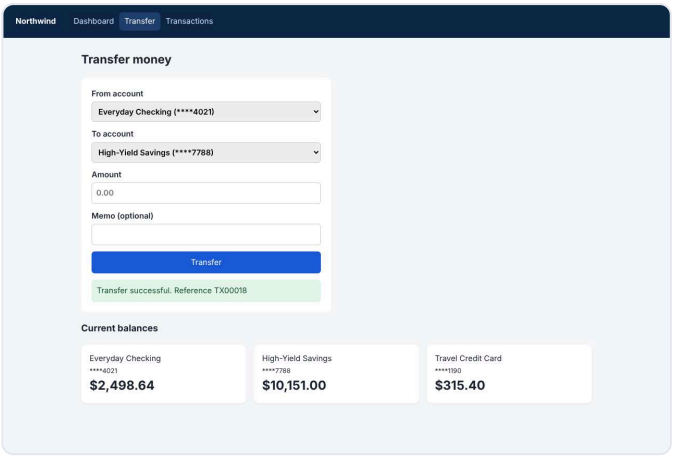
after step 0: Open http://localhost:8080 and click 'Enter Demo'



after step 1: Click 'Transfer' in the top navigation



after step 7: Click the 'Transfer' submit button



after step 9: Read 'testid=balance-sav' as sav_after

HIGH BUG-04 · Not met: Overdraft is rejected with Insufficient funds

SC-07-a · negative scenario · requirements REQ-07 · reproduced on 2 attempt(s)

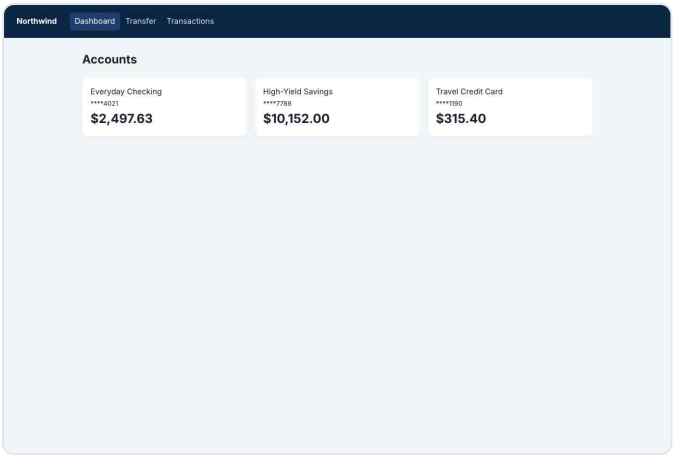
Steps to reproduce

1. Open http://localhost:8080 and click 'Enter Demo'
2. Click 'Transfer' in the top navigation
3. Read 'testid=balance-chk' as chk_before → read 2497.63
4. Read 'testid=balance-sav' as sav_before → read 10152.0
5. Select 'Everyday Checking (****4021)' in 'label=From account'
6. Select 'High-Yield Savings (****7788)' in 'label=To account'
7. Enter '999999' into 'label=Amount'
8. Click the 'Transfer' submit button
9. Read 'testid=balance-chk' as chk_after → read -997501.37
10. Read 'testid=balance-sav' as sav_after → read 1010151.0

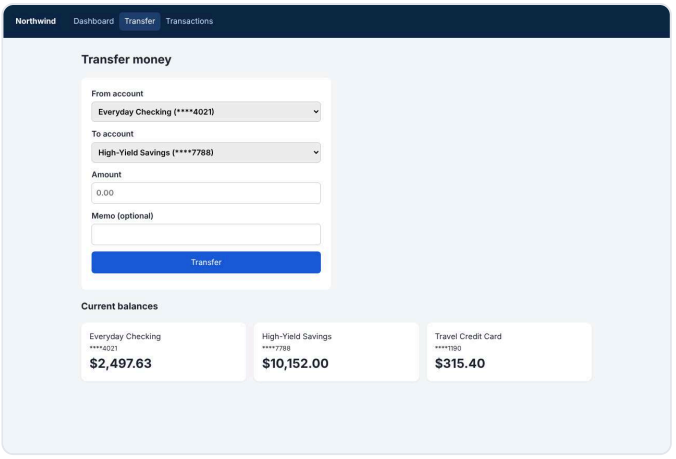
Expected	'Insufficient funds' shown, balances unchanged.
Observed	no error or validation message visible; chk_after=-997501.37, chk_before=2497.63; sav_after=1010151.0, sav_before=10152.0
Failed check	Insufficient funds message expected `a visible validation / error message mentioning 'Insufficient funds'` observed `no error or validation message visible`
Failed check	Checking balance unchanged expected `{chk_after} == {chk_before}` observed `chk_after=-997501.37, chk_before=2497.63`
Failed check	Savings balance unchanged expected `{sav_after} == {sav_before}` observed `sav_after=1010151.0, sav_before=10152.0`
Captured values	`{'chk_before': 2497.63, 'sav_before': 10152.0, 'chk_after': -997501.37, 'sav_after': 1010151.0}`

API evidence (1)

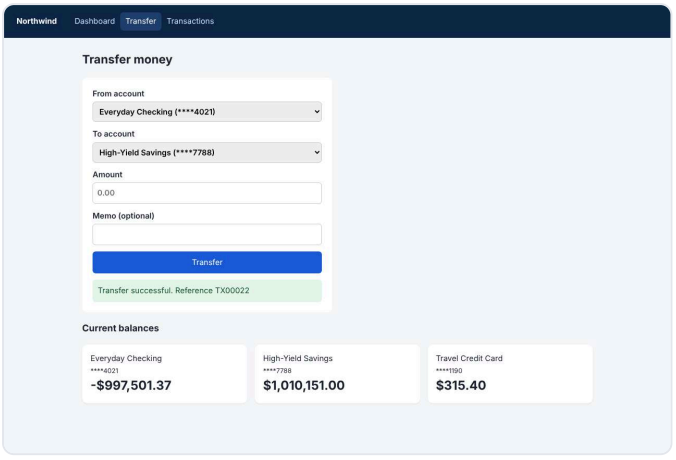
```
POST http://localhost:8080/api/transfer → 200
request: {"from":"chk","to":"sav","amount":"999999","memo":""}
response: {"ok": true, "reference": "TX00022"}
```



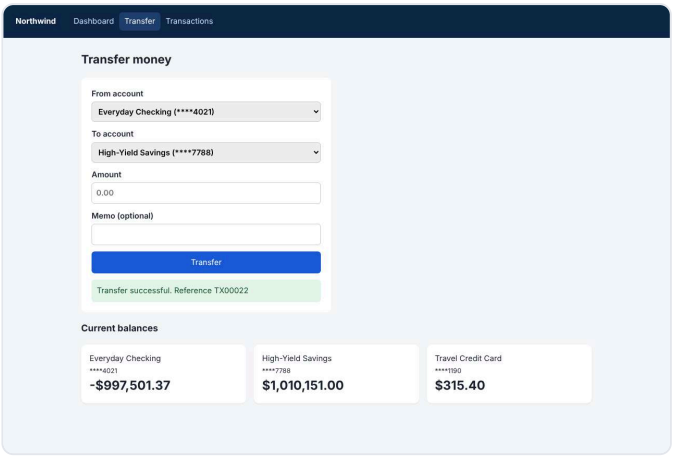
after step 0: Open http://localhost:8080 and click 'Enter Demo'



after step 1: Click 'Transfer' in the top navigation



after step 7: Click the 'Transfer' submit button



after step 9: Read 'testid=balance-sav' as sav_after

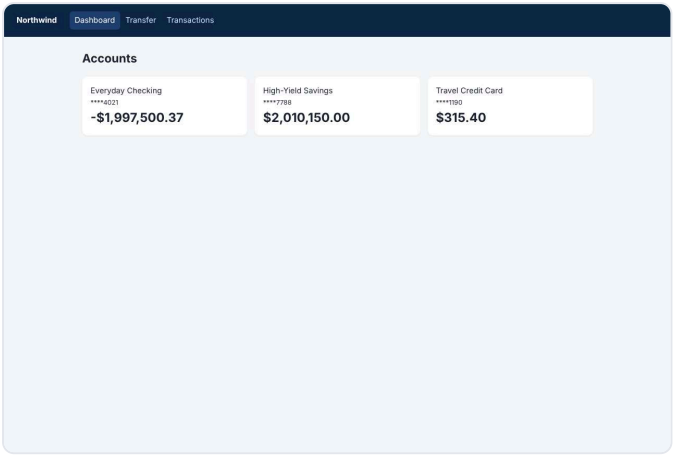
HIGH **BUG-05 · Not met: Debit filter shows only debits**

SC-11-a · normal scenario · requirements REQ-11 · reproduced on 2 attempt(s)

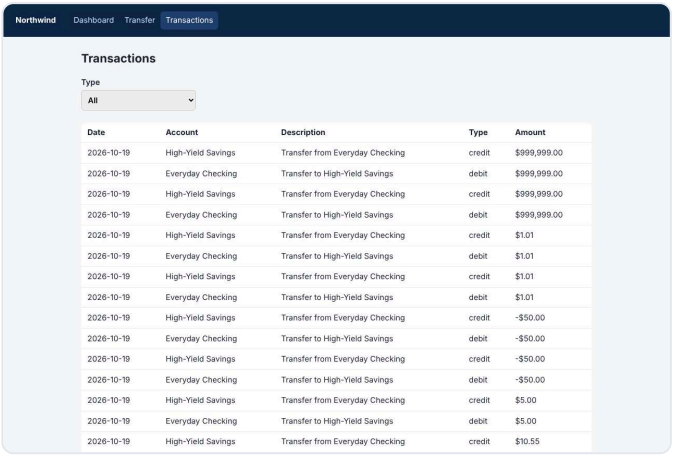
Steps to reproduce

1. Open http://localhost:8080 and click 'Enter Demo'
2. Click 'Transactions' in the top navigation
3. Select 'Debit' in 'label=Type'

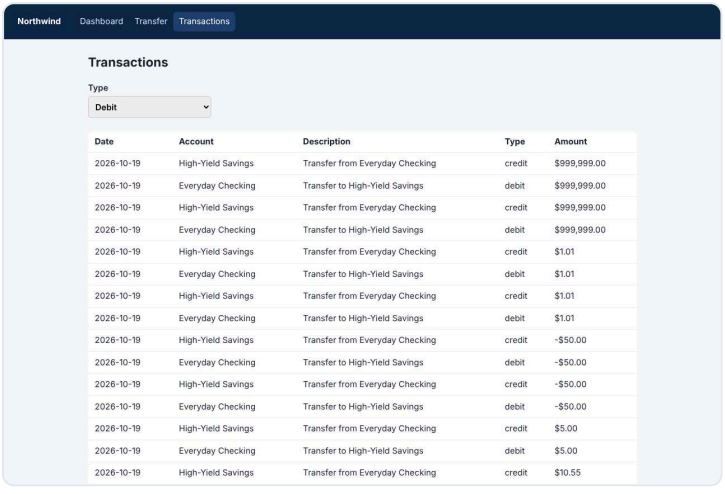
Expected	Only debit rows.
Observed	present on page
Failed check	No credit rows in table expected `text 'credit' not visible` observed `present on page`



after step 0: Open http://localhost:8080 and click 'Enter Demo'



after step 1: Click 'Transactions' in the top navigation



after step 2: Select 'Debit' in 'label=Type'

3 · Unverified (0)

Scenarios the agent could not decide with evidence. They are not counted as defects, to avoid false positives.

4 · All scenarios

ID	TYPE	SCENARIO	VERDICT	CHECKS	HEALING	TIME
SC-01-a	normal	**Enter Demo lands on the account dashboard** Entry flow is the gateway to every other test 1. Open http://localhost:8080 and click 'Enter Demo' ✓ Accounts heading is shown — visible (via text) ✓ Checking account is listed — present on page ✓ URL points at the dashboard — http://localhost:8080/#/dashboard	PASSED	3/3	—	1.65s
SC-02-a	normal	**Balances use $ and two decimals** Formatting bugs are common with floats 1. Open http://localhost:8080 and click 'Enter Demo' ✓ Checking balance format — '$2,450.75' ✓ Savings balance format — '$10,200.00' ✓ Masked number shown — present on page	PASSED	3/3	—	1.61s
SC-03-a	normal	**Navigation offers the three sections** Missing navigation blocks whole features 1. Open http://localhost:8080 and click 'Enter Demo' ✓ Dashboard nav — visible (via role) ✓ Transfer nav — visible (via role) ✓ Transactions nav — visible (via role)	PASSED	3/3	—	1.62s
SC-04-a	normal	**Whole-dollar transfer updates both balances** A success toast does not prove the ledger moved 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transfer' in the top navigation 3. Read 'testid=balance-chk' as chk_before → read 2450.75 4. Read 'testid=balance-sav' as sav_before → read 10200.0 5. Select 'Everyday Checking (****4021)' in 'label=From account' 6. Select 'High-Yield Savings (****7788)' in 'label=To account' 7. Enter '25.00' into 'label=Amount'	PASSED	4/4	role:button	6.08s

ID	TYPE	SCENARIO	VERDICT	CHECKS	HEALING	TIME
		8. Click the 'Transfer' submit button 9. Read 'testid=balance-chk' as chk_after → read 2425.75 10. Read 'testid=balance-sav' as sav_after → read 10225.0 ✓ Source decreased by 25.00 — chk_after=2425.75, chk_before=2450.75 ✓ Destination increased by 25.00 — sav_after=10225.0, sav_before=10200.0 ✓ Confirmation with reference — present on page ✓ No server errors — none				
SC-04-b	edge	**Cents are transferred exactly (10.55)** Float/int handling often drops cents 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transfer' in the top navigation 3. Read 'testid=balance-chk' as chk_before → read 2425.75 4. Read 'testid=balance-sav' as sav_before → read 10225.0 5. Select 'Everyday Checking (****4021)' in 'label=From account' 6. Select 'High-Yield Savings (****7788)' in 'label=To account' 7. Enter '10.55' into 'label=Amount' 8. Click the 'Transfer' submit button 9. Read 'testid=balance-chk' as chk_after → read 2415.2 10. Read 'testid=balance-sav' as sav_after → read 10235.0 ✓ Source decreased by 10.55 — chk_after=2415.2, chk_before=2425.75 × Destination increased by 10.55 — sav_after=10235.0, sav_before=10225.0	FAILED	1/2	role:button	12.33s
SC-05-a	normal	**Transfer adds a debit and a credit row** Ledger and balances can drift apart 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transactions' in the top navigation 3. Read 'css=#tx-table tbody tr' as rows_before → read 11 4. Click 'Transfer' in the top navigation 5. Read 'testid=balance-chk' as chk_before → read 2404.65 6. Read 'testid=balance-sav' as sav_before → read 10245.0	PASSED	3/3	role:button	7.52s

ID	TYPE	SCENARIO	VERDICT	CHECKS	HEALING	TIME
		7. Select 'Everyday Checking (****4021)' in 'label=From account' 8. Select 'High-Yield Savings (****7788)' in 'label=To account' 9. Enter '5.00' into 'label=Amount' 10. Click the 'Transfer' submit button 11. Read 'testid=balance-chk' as chk_after → read 2399.65 12. Read 'testid=balance-sav' as sav_after → read 10250.0 13. Click 'Transactions' in the top navigation 14. Read 'css=#tx-table tbody tr' as rows_after → read 13 ✔ Exactly two new rows — rows_after=13, rows_before=11 ✔ Debit row names destination — present on page ✔ Credit row names source — present on page				
SC-06-a	negative	**Negative amount is rejected** Sign handling lets users pull money backwards 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transfer' in the top navigation 3. Read 'testid=balance-chk' as chk_before → read 2399.65 4. Read 'testid=balance-sav' as sav_before → read 10250.0 5. Select 'Everyday Checking (****4021)' in 'label=From account' 6. Select 'High-Yield Savings (****7788)' in 'label=To account' 7. Enter '-50' into 'label=Amount' 8. Click the 'Transfer' submit button 9. Read 'testid=balance-chk' as chk_after → read 2449.65 10. Read 'testid=balance-sav' as sav_after → read 10200.0 ✘ An error message is shown — no error or validation message visible ✘ Checking balance unchanged — chk_after=2449.65, chk_before=2399.65 ✘ Savings balance unchanged — sav_after=10200.0, sav_before=10250.0	FAILED	0/3	role:button	11.81s
SC-06-b	negative	**Non-numeric amount is rejected** Parsing errors 1. Open http://localhost:8080 and click 'Enter Demo'	PASSED	3/3	role:button	5.88s

ID	TYPE	SCENARIO	VERDICT	CHECKS	HEALING	TIME
		2. Click 'Transfer' in the top navigation 3. Read 'testid=balance-chk' as chk_before → read 2499.65 4. Read 'testid=balance-sav' as sav_before → read 10150.0 5. Select 'Everyday Checking (****4021)' in 'label=From account' 6. Select 'High-Yield Savings (****7788)' in 'label=To account' 7. Enter 'abc' into 'label=Amount' 8. Click the 'Transfer' submit button 9. Read 'testid=balance-chk' as chk_after → read 2499.65 10. Read 'testid=balance-sav' as sav_after → read 10150.0 ✓ An error message is shown — Enter a valid amount ✓ Checking balance unchanged — chk_after=2499.65, chk_before=2499.65 ✓ Savings balance unchanged — sav_after=10150.0, sav_before=10150.0				
SC-06-c	edge	**Zero amount is rejected** Boundary at 0 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transfer' in the top navigation 3. Read 'testid=balance-chk' as chk_before → read 2499.65 4. Read 'testid=balance-sav' as sav_before → read 10150.0 5. Select 'Everyday Checking (****4021)' in 'label=From account' 6. Select 'High-Yield Savings (****7788)' in 'label=To account' 7. Enter '0' into 'label=Amount' 8. Click the 'Transfer' submit button 9. Read 'testid=balance-chk' as chk_after → read 2499.65 10. Read 'testid=balance-sav' as sav_after → read 10150.0 ✓ An error message is shown — Amount must be greater than zero ✓ Checking balance unchanged — chk_after=2499.65, chk_before=2499.65 ✓ Savings balance unchanged — sav_after=10150.0, sav_before=10150.0	PASSED	3/3	role:button	5.92s
SC-06-d	edge	**Three decimal places are rejected (1.005)** PRD caps precision at two decimals	FAILED	0/3	role:button	11.86s

ID	TYPE	SCENARIO	VERDICT	CHECKS	HEALING	TIME
		1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transfer' in the top navigation 3. Read 'testid=balance-chk' as chk_before → read 2499.65 4. Read 'testid=balance-sav' as sav_before → read 10150.0 5. Select 'Everyday Checking (****4021)' in 'label=From account' 6. Select 'High-Yield Savings (****7788)' in 'label=To account' 7. Enter '1.005' into 'label=Amount' 8. Click the 'Transfer' submit button 9. Read 'testid=balance-chk' as chk_after → read 2498.64 10. Read 'testid=balance-sav' as sav_after → read 10151.0 × An error message is shown — no error or validation message visible × Checking balance unchanged — chk_after=2498.64, chk_before=2499.65 × Savings balance unchanged — sav_after=10151.0, sav_before=10150.0				
SC-07-a	negative	**Overdraft is rejected with Insufficient funds** Money integrity 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transfer' in the top navigation 3. Read 'testid=balance-chk' as chk_before → read 2497.63 4. Read 'testid=balance-sav' as sav_before → read 10152.0 5. Select 'Everyday Checking (****4021)' in 'label=From account' 6. Select 'High-Yield Savings (****7788)' in 'label=To account' 7. Enter '999999' into 'label=Amount' 8. Click the 'Transfer' submit button 9. Read 'testid=balance-chk' as chk_after → read -997501.37 10. Read 'testid=balance-sav' as sav_after → read 1010151.0 × Insufficient funds message — no error or validation message visible × Checking balance unchanged — chk_after=-997501.37, chk_before=2497.63 × Savings balance unchanged — sav_after=1010151.0, sav_before=10152.0	FAILED	0/3	role:button	11.97s

ID	TYPE	SCENARIO	VERDICT	CHECKS	HEALING	TIME
SC-08-a	negative	**Same source and destination is rejected** Self-transfer loops 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transfer' in the top navigation 3. Read 'testid=balance-chk' as chk_before → read -1997500.37 4. Read 'testid=balance-sav' as sav_before → read 2010150.0 5. Select 'Everyday Checking (****4021)' in 'label=From account' 6. Select 'Everyday Checking (****4021)' in 'label=To account' 7. Enter '5' into 'label=Amount' 8. Click the 'Transfer' submit button 9. Read 'testid=balance-chk' as chk_after → read -1997500.37 10. Read 'testid=balance-sav' as sav_after → read 2010150.0 ✔ An error message is shown — Source and destination must differ ✔ Checking unchanged — chk_after=-1997500.37, chk_before=-1997500.37	PASSED	2/2	role:button	5.98s
SC-09-a	negative	**Empty amount shows validation** Required-field handling 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transfer' in the top navigation 3. Read 'testid=balance-chk' as chk_before → read -1997500.37 4. Read 'testid=balance-sav' as sav_before → read 2010150.0 5. Select 'Everyday Checking (****4021)' in 'label=From account' 6. Select 'High-Yield Savings (****7788)' in 'label=To account' 7. Clear 'label=Amount' 8. Click the 'Transfer' submit button 9. Read 'testid=balance-chk' as chk_after → read -1997500.37 10. Read 'testid=balance-sav' as sav_after → read 2010150.0 ✔ Validation message shown — Enter an amount ✔ Checking balance unchanged — chk_after=-1997500.37, chk_before=-1997500.37	PASSED	3/3	role:button	5.47s

ID	TYPE	SCENARIO	VERDICT	CHECKS	HEALING	TIME
		✓ Savings balance unchanged — sav_after=2010150.0, sav_before=2010150.0				
SC-10-a	normal	**Newest transaction appears first** Sort order 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transactions' in the top navigation 3. Read 'css=#tx-table tbody tr:first-child td:first-child' as first_date → read '2026-10-19' 4. Read 'css=#tx-table tbody tr:last-child td:first-child' as last_date → read '2026-10-01' ✓ First row is not older than last row — first_date=2026-10-19, last_date=2026-10-01 ✓ Amount column present — visible (via text)	PASSED	2/2	role:button	2.4s
SC-11-a	normal	**Debit filter shows only debits** Filter logic 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transactions' in the top navigation 3. Select 'Debit' in 'label=Type' × No credit rows in table — present on page ✓ Debit rows present — present on page	FAILED	1/2	role:button	6.21s
SC-11-b	normal	**Credit filter shows only credits** Filter logic 1. Open http://localhost:8080 and click 'Enter Demo' 2. Click 'Transactions' in the top navigation 3. Select 'Credit' in 'label=Type' ✓ No debit rows in table — not found on page ✓ Credit rows present — present on page	PASSED	2/2	role:button	3.12s

5 · Cost breakdown

MODEL	CALLS (CACHED)	INPUT TOKENS	CACHED INPUT	OUTPUT TOKENS	RATE $/1M (IN / CACHED / OUT)	COST
No model calls in this run (replayed a saved plan; all checks deterministic).						

STAGE	WHAT THE MODEL DOES	CALLS (CACHED)	COST
`exploration, execution, oracles`	Playwright + deterministic checks	0	$0.0000

Token counts come from the API's own usage fields (including image tokens and prompt-cache hits). Prices from pricing.json — verify against the provider's current price list. Repeat runs of an unchanged plan are served from the local response cache at $0.

6 · Method

1. **Explore** (no LLM): enter the demo, discover views, record fields, buttons, tables and stable selectors.
2. **Plan** (plan model): extract explicit + implied requirements from the PRD, then design normal / negative / edge scenarios grounded in the explored site map.
3. **Execute**: one fresh browser context per scenario; self-healing element resolution (test id → label → placeholder → ARIA role → text → fast-model pick from live DOM); network-idle + DOM-quiet waits instead of sleeps.
4. **Verify**: deterministic oracles over captured before/after values, DOM state, validation messages and HTTP traffic. A success toast is never accepted as proof.
5. **Confirm**: failures re-run once in a clean context (flake filter), then reviewed by the judge model, which can downgrade test-design mistakes to UNVERIFIED.